

Reasoning and Planning for Multimodal Large Language Models: A Multilingual and Cross-Domain Exploration

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Abstract

Recent advancements in Multimodal Large Language Models (MLLMs), coupled with the progress of reinforcement learning, have substantially enhanced reasoning and decision-making across modalities, including text, vision, audio, and video. This tutorial introduces the fundamental principles, methodologies, and practical applications of MLLM reasoning, with a particular emphasis on strengthening reasoning capabilities in multilingual and cross-domain settings. We further discuss the key challenges and limitations of current multimodal reasoning approaches, as well as future directions for advancing the field. By highlighting how MLLMs support enhanced reasoning and planning in cross-lingual and cross-domain contexts, this session aims to equip researchers and practitioners with the conceptual foundations and practical tools needed to effectively integrate MLLM reasoning into their work.

CCS Concepts

• Computing methodologies → Natural language generation.

Keywords

Multimodal Reasoning, Instruction Tuning, Large Language Models

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1 Proposer/Speaker Information

The proposers and speakers are Ms. Sarmistha Das, Mr. Akash Ghosh, and Dr. Sriparna Saha (IIT Patna), with industrial collaborators Dr. Koustava Goswami and Dr. Joseph K. J. (Adobe Research, India). Sarmistha Das is a PhD scholar at IIT Patna working on multimodal AI and FinTech, with publications in WACV, PAKDD, ECAI, ACM-MM, TCSS and CIKM. Akash Ghosh, also a PhD scholar

at IIT Patna, focuses on multimodal and multilingual NLP, with publications in AAAI, ACL, EMNLP, and ECIR. Dr. Sriparna Saha, Associate Professor at IIT Patna (h5-index:51, 13,015+ citations), has 400+ papers in ACL, AAAI, ACM-MM, SIGIR, EMNLP, CIKM, etc., and serves as Associate/Area Editor for several IEEE/ACM journals. Dr. Koustava Goswami (PhD, University of Galway) is a Research Scientist at Adobe with 35+ publications in ACL, EMNLP, ICCV, ECCV, and WACV. Dr. Joseph K. J. (PhD, IIT Hyderabad) is a Research Scientist at Adobe with publications in CVPR, ICCV, ECCV, TPAMI, NeurIPS, and WACV.

2 Breakdown of Tutorial

Multimodal Large Language Models (MLLMs) unify text, vision, and audio for enhanced reasoning. Key aspects include architecture, modality interactions, and instruction tuning. Reasoning is improved via exemplar generation and model interaction. Evaluation spans closed- and open-set tasks in vision, language, and cognition. Applications range from embodied AI to multilingual tutoring. Domain-specific reasoning highlights cross-lingual adaptability and language bias. Future directions involve reducing hallucination, strengthening multimodal alignment, and improving efficiency.

3 Motivation

Effective reasoning across multiple modalities is crucial for AI systems to navigate complex environments and make informed decisions. Traditional unimodal approaches often fall short when dealing with ambiguous or incomplete information, limiting their practical applicability. In visual crosswords, an MLLM can align image cues (e.g., a banana) with textual clues (“yellow, long fruit”), demonstrating the power of multimodal reasoning for accurate inference.

4 Course Description

The tentative name of the speaker is provided with each of the module. We followed the following acronyms: Sarmistha Das (SD), Akash Ghosh (AG), Sriparna Saha (SS), Koustava Goswami (KG), Joseph K J (JKJ).

4.1 Preliminaries

[SS]

- **Multimodal Large Language Models:** MLLMs combine a visual encoder, a language model, and a connector. Key differences arise in three areas: Training Data, Vision-Language Interaction, and Connector Design.

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- **Reasoning and Planning in MLLMs:** Reasoning in MLLMs involves drawing logical conclusions by analyzing and interpreting multimodal data, while planning refers to generating structured action sequences to achieve specific goals. By integrating multiple modalities, MLLMs enhance inference capabilities, enabling more context-aware decision-making and improving performance in complex, real-world tasks.
- **Instruction Tuning for MLLMs:** MLLMs improve reasoning via instruction tuning on curated instruction-response pairs. In multimodal settings, modality-specific data enhances context awareness and response quality.

4.2 Different Thrusts to enhance Multimodal Reasoning and Planning [JKJ]

- **Instruction Tuning for In-Context Learning (ICL):** Strong analogical reasoning is crucial for ICL, enabling LLMs to adapt to new tasks with few examples and no weight updates [1].
- **Multimodal Prompting by Representation Learning:** Multimodal prompting risks catastrophic forgetting due to data limits. Aligning non-linguistic inputs with LLMs without altering the base model is challenging. Solutions include linear projection and vision encoders.
- **Multimodal Prompting by Exemplar Generation:** In-context examples enhance analogical reasoning in LLMs, but multimodal prompting is less explored due to the scarcity of examples. Automatic exemplar generation [3] extracts answer candidates from the generated captions.
- **Multimodal Prompting by Model Interaction:** Integrating foundation models across modalities is challenging and costly. Multimodal prompting enables zero-shot learning using language prompts and structured interactions without additional training.
- **Adaptive and Structured Planning:** Multimodal Large Language Models (MLLMs) can enhance their planning capabilities by integrating cross-modal alignment (synchronizing different data types for coherent decision-making) and hierarchical planning (breaking complex tasks into structured action sequences) [5]. Additionally, dynamic context adaptation enables models to adjust their plans based on evolving real-world inputs, improving responsiveness in dynamic environments.

4.3 Benchmarks and Evaluation [AG]

This tutorial introduces a comprehensive evaluation framework for Multimodal Large Language Models (MLLMs) through robust benchmarks assessing reasoning, planning, and generalization across modalities. It emphasizes closed-set (predefined choices) and open-set (free-form) evaluation, focusing on three pillars: vision-based reasoning (visual analysis), linguistic reasoning (textual coherence and comprehension), and multimodal reasoning (cross-modal integration for decision-making). Participants will gain practical insights into benchmarking, interpreting, and improving MLLM performance in real-world applications.

4.4 Applications [SS + JKJ]

- **MLLMs for Embodied AI:**
 - Enhanced task decomposition and learning through pre-trained MLLMs with commonsense reasoning and fine-tuned VLMs on small datasets.

- Adaptive multimodal perception and generalization by integrating visual, textual, and sensor data for scene understanding and cross-task reasoning.

• MLLMs as Tools:

- Compositional visual reasoning for tasks like detection, recognition, and merging (e.g., face swapping).
- Multimodal dialogue reasoning to improve conversations by combining visual inputs with textual reasoning.
- Integration of modalities beyond vision, using LLMs for reasoning over tables, documents, and robotics data.

4.5 Reasoning of MLLM in Various Domains and Languages [SD+AG]

Multimodal Large Language Models (MLLMs) integrate text, vision, and audio to enhance reasoning across domains [5]. They assist in medical diagnosis [4], multilingual tutoring, scientific analysis [6], contract review, fraud detection, and robotics [2]. MLLMs also generate creative content but face challenges in multilingual alignment due to high-resource language bias.

4.6 Scope for Future Work [KG]

Future MLLM research should address hallucinations from complex multimodal inputs, improve architectures, optimize training efficiency, and scale models with data volume. Enhancing long-context support will aid in handling extended documents and videos. Better instruction fine-tuning datasets will strengthen multimodal reasoning, and reinforcement learning for multimodal reasoning requires scalable preference data and improved agents.

5 Audience Information

The tutorial's content ranges from basic to advanced, making it suitable for beginners to expert learners with varying levels of understanding. It is designed to cater to the needs of both academic and industry professionals, including undergraduate and master students, Ph.D. students, and personnel working in the industry.

6 Supporting Material

The tutorial provides the following material: (1) lecture video recording, (2) annotated slides, (3) assignments, exams and projects from related courses taught, (4) an extended bibliography, and (5) a compendium of public software and data sets. The resources are available at: <https://github.com/sarmistha-D/ACM-Tutorial-MLLM>

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